

CSE 719: Distributed & Cloud Computing

Lecture 2 — Cloud Computing

Model Answers: 2020, 2021, 2022 & 2024 Papers

University of Chittagong — 8th Sem B.Sc. (Engg.) — Sources: Lecture-02 + Unit 3/4 Readings

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2020 Examination (CSE-813, 52.5 marks)

*Note: 2020 Q1(a) “Define distributed and cloud systems” is answered in **Fundamentals_Solutions.pdf** (shared question, see that file).*

Q8(a) — Define cloud computing; advantages and disadvantages [2 marks]

Define cloud computing with its advantages and disadvantages.

Definition (NIST): Cloud computing is a model for enabling **convenient, on-demand network access to a shared pool of configurable computing resources** (networks, servers, storage, applications, services) that can be **rapidly provisioned and released** with minimal management effort or service-provider interaction.

Advantages:

- **Scalability / Flexibility** — instantly scale resources up or down; access from any internet-connected device.
- **Cost reduction** — pay-as-you-go; no upfront capital expenditure on hardware.
- **Maintenance offloaded** — vendor handles patches, updates, monitoring, and backups.
- **High availability** — provider manages fault tolerance; 99.9%+ SLA uptime guaranteed.

Disadvantages:

- **Security & Privacy** — sensitive data resides on third-party shared infrastructure.
- **Vendor lock-in** — proprietary APIs make migration to another provider costly.
- **Network dependency** — no internet $\Rightarrow$ no access to applications or data.
- **Data migration** — moving large datasets in/out of a provider is slow and expensive.

Q8(b) — E-commerce website: cloud services and deployment models [4 marks]

What cloud services and deployment models would you choose for building an e-commerce website? Justify your choices.

Cloud Services (by layer):

Need	Service Model	Justification
Web app hosting	PaaS	Runtime + middleware pre-configured; no OS management; auto-scaling built in (e.g. Google App Engine, Heroku).
Database / Storage	IaaS	Full control over DB config, replication, and storage volumes (e.g. AWS EC2 + EBS).
Payment / CRM / Email	SaaS	Off-the-shelf, no custom build — Stripe (payments), Salesforce (CRM), SendGrid (email).

Deployment Model:

- **Public cloud** (e.g. AWS, Azure) for product catalog, storefront, CDN — globally distributed, cost-effective, handles traffic spikes.
- **Private cloud** for payment processing and customer PII — required for regulatory compliance (PCI-DSS, GDPR).
- **Hybrid cloud** overall: routine traffic on public; sensitive data on private; **cloud bursting** to public during sales peaks.

Justification: Hybrid maximises cost efficiency while satisfying compliance. PaaS accelerates development; SaaS avoids reinventing commodity services.

Q8(c) — Cloud aspects for prognostic health management [2.75 marks]

What aspects of cloud computing are useful for prognostic health management applications?

- **Elastic scalability** — health sensor data volume is unpredictable; cloud elastically provisions compute for ML inference and analytics bursts on demand.
- **Storage elasticity** — continuous streams of patient/sensor data require virtually unlimited storage (e.g. Amazon S3, Azure Blob); no pre-purchase.
- **High availability / Reliability** — provider-managed fault tolerance ensures 24/7 uptime; critical for real-time health monitoring alerts.

- **Pay-per-use** — health analytics is batch-heavy; pay only during computation phases, not for idle hardware.
- **Security controls** — encryption at rest/in transit, role-based access control, and audit logs required under HIPAA/health regulations.

2021 Examination (CSE 813, 52.5 marks)

Q3(a) — E-commerce: cloud services and deployment models [2.75 marks]

Which cloud services and deployment models would you adopt for building an e-commerce website?

Cloud services:

- **IaaS** — virtual servers for databases; object storage for product images.
- **PaaS** — web application hosting platform: runtime, middleware, and auto-scaling.
- **SaaS** — payment gateway, email, CRM (no custom build needed).

Deployment model:

- **Hybrid** = Public cloud for storefront and CDN (scalable, low cost) + Private cloud for payment/customer PII (regulatory compliance).
- **Cloud bursting**: overflow peak-season traffic to public cloud automatically.

Q4(c) — Phases in cloud architecture

[1 marks]

What are the phases involved in cloud architecture?

Cloud architecture is built in three successive phases:

1. **Infrastructure phase (IaaS)** — physical hardware (servers, storage, network) is abstracted via **virtualization (hypervisor)** into on-demand virtual resources.
2. **Platform phase (PaaS)** — runtime environment, middleware, databases, and dev/deployment tools are provisioned on top of the virtual infrastructure.

3. **Application phase (SaaS)** — end-user software is delivered over the internet; fully managed by the provider with no user infrastructure responsibility.

Q5(a) — On-demand functionality in cloud

[1.75 marks]

What is on-demand functionality? How is it provided in cloud computing?

On-demand: Computing resources (compute, storage, bandwidth) are **provisioned automatically and immediately** as the user requests them, **without any human intervention** from the provider's side.

How it is provided:

1. Provider maintains **large shared resource pools** (CPU, RAM, disk, bandwidth).
2. User requests resources via a **self-service portal or API** (no phone call to provider needed).
3. The **hypervisor / orchestration layer** allocates a VM or container within seconds.
4. **Auto-scaling** monitors load and adds/removes instances dynamically.
5. **Metering** tracks consumption; billing reflects actual usage only.

Key idea: the user perceives an **illusion of infinite resources** available instantly on demand.

Q5(b) — What is a cloud? Large-scale cloud platforms

[1 marks]

What is a cloud? Name some platforms used for large-scale cloud computing.

Cloud (Vaquero et al.): “A large pool of easily usable and accessible **virtualised resources** (hardware, development platforms, services), dynamically reconfigured to adjust to a variable load, allowing for optimum resource utilisation via a **pay-per-use** model where guarantees are offered by the provider via SLAs.”

Large-scale cloud platforms:

- **Amazon EC2 / S3 (AWS)** — dominant public IaaS; virtual machines + object storage.
- **Microsoft Azure** — enterprise-focused public cloud; strong hybrid integration.

- **Google Cloud / GCE** — strong in big data analytics and Kubernetes.
- **EUCALYPTUS** — open-source **private** cloud platform, AWS-compatible API.

Q5(c) — Layers in cloud; cloud vs. distributed computing [3 marks]

Explain the layers in cloud computing and how they work. Differentiate between cloud and distributed computing.

Layers in cloud (bottom to top):

Layer	Model	What it provides
1. Physical	Datacenter	Servers, switches, power, cooling — owned by provider.
2. Virtualization	Hypervisor	Abstracts physical hardware into isolated VMs.
3. Infrastructure	IaaS	On-demand VMs, virtual storage, virtual networking.
4. Platform	PaaS	Runtime, DB, dev tools on top of IaaS.
5. Application	SaaS	End-user apps delivered via browser; zero local install.

How it works: User code runs in SaaS; SaaS is built on PaaS (execution environment); PaaS runs on IaaS (virtual servers); IaaS sits on the physical datacenter.

Cloud vs. Distributed Computing:

Criterion	Distributed Computing	Cloud Computing
Scope	General model: independent nodes cooperating over a network	Specific implementation delivered as a metered utility service
Business model	None inherent	Pay-per-use (OPEX, not CAPEX)
Provisioning	Static / pre-allocated by admin	Elastic, self-service, on-demand
Management	User/admin manages nodes	Provider manages all infrastructure
Multi-tenancy	Rare	Core feature (shared physical resources)
Examples	DNS, P2P networks, MPI cluster	AWS EC2, Azure, Google Cloud

Key: Every cloud is a distributed system, but not every distributed system is a cloud.

Q5(d) — EUCALYPTUS: what it is, why used, deployment models [3 marks]

What is EUCALYPTUS? Why is it used? Explain the cloud deployment models.

EUCALYPTUS (*Elastic Utility Computing Architecture for Linking Your Programs To Useful Systems*):

- Open-source software framework for building **private cloud environments** on an organisation's existing on-premises hardware.
- Implements an **AWS-compatible API**: tools and scripts written for Amazon EC2 work on EUCALYPTUS without modification.
- Provides IaaS capabilities: VM provisioning, virtual networking, elastic storage — all behind the organisation's firewall.

Why used:

- Organisations want cloud elasticity and self-service but **cannot put data in a public cloud** (security policy, data sovereignty regulations, low-latency requirements).
- Virtualises in-house servers to reduce idle capacity and hardware sprawl.
- AWS-compatible API provides a **future migration path** to public cloud.

Cloud Deployment Models:

Model	Description
Public	Infrastructure owned by a provider (IBM, Google, Amazon); open to paying customers over the internet.
Private	Infrastructure dedicated to a single organisation; may be on-premises (EUCALYPTUS) or hosted exclusively.
Hybrid	Combination of public + private; private for sensitive/regulated workloads, public for elastic bursting.
Community	Shared by a consortium of organisations with common concerns (e.g. research universities, government agencies).

2022 Examination (CSE 513, 54 marks)

Q1(a) — Define cloud computing; horizontal vs. vertical scaling [5 marks]

Define cloud computing. Explain horizontal and vertical scaling with suitable scenarios.

Definition (NIST). Cloud computing is a model for enabling **convenient, on-demand network access to a shared pool of configurable computing resources** that can be **rapidly provisioned and released** with minimal management effort or service provider interaction.

Scaling strategies:

Criterion	Vertical Scaling (Scale Up)	Horizontal Scaling (Scale Out)
Method	Add more CPU/RAM/disk to the same machine	Add more machines to the pool
Upper limit	Hardware ceiling of one server	Virtually unlimited (commodity nodes)
Downtime	Often requires restart	None (add/remove nodes live)
App changes	Usually none	May need load balancer, stateless design
Fault tolerance	Single point of failure	No single point of failure
Cost	Expensive (enterprise hardware)	Cheap (commodity servers)

Scenarios:

- **Vertical** — a legacy monolithic relational database (Oracle) that cannot be sharded: give it more RAM/CPU cores. Also suitable as a short-term emergency fix.
- **Horizontal** — a stateless web/application server tier: add instances behind a load balancer. Cloud auto-scaling groups exploit this model for variable traffic.

Cloud preference: Horizontal scaling aligns with cloud economics — use many cheap VMs; add and remove them elastically.

Q1(b) — E-commerce: cloud services and deployment model [2 marks]

Which cloud services and deployment model would you suggest for an e-commerce website?

- **IaaS** for database VMs and object storage (product images, order logs).
- **PaaS** for web application tier — auto-scaled runtime, no OS management.
- **SaaS** for payment gateway, email service, and analytics.
- **Hybrid deployment:** public cloud for storefront and CDN (scalable, cheap); private cloud for payment/customer PII (compliance); cloud burst to public at peak sales.

Q1(c) — What is multi-tenancy? [2 marks]

What is multi-tenancy in cloud computing?

Multi-tenancy: Multiple customers (*tenants*) share the **same physical infrastructure and software instances**, while each tenant's data and configuration remain **logically isolated** from every other tenant.

Why it matters:

- **Economic model:** provider spreads fixed infrastructure costs across many tenants $\Rightarrow$ lower per-customer price.
- **High server utilisation:** shared servers are rarely idle; CPU cycles are sold continuously.

Risks:

- **Data leakage** between tenants if VM isolation is misconfigured.

- **Noisy neighbour** effect: one tenant's heavy workload degrades another's performance on the same physical host.

SaaS maturity (Levels 1–4): Level 1 = separate instance per tenant (weakest, no sharing). Level 4 = single instance, single database, configurable per-tenant views (most efficient multi-tenancy).

Q2(a) — EC2 VMs vs. physical machines: three benefits to Amazon [3 marks]

Compare EC2 virtual machines with physical machines. State three benefits Amazon gains by offering VMs instead of leasing physical machines. What motivated Amazon to start EC2?

Three benefits to Amazon of VMs over physical machine leasing:

1. **Higher server utilisation** — multiple VMs packed on one physical server raise utilisation from ~10–20% to >80%. More revenue per rack, same power and cooling cost.
2. **Flexible, fine-grained billing** — sell fractional capacity (micro, small, large, GPU instances) at different price points; maximise yield from each physical machine. Physical leasing is coarse-grained (whole server).
3. **Strong isolation and safe multi-tenancy** — hypervisor provides hardware-level memory and I/O isolation between tenants; one customer's VM cannot read another's memory. Enables renting the same physical server to many customers simultaneously.

Motivation for EC2: Amazon sized its datacenters for **peak holiday shopping traffic**. During 99% of the year that capacity was idle. EC2 was launched in 2006 to **monetise excess capacity** — what began as internal infrastructure became the world's dominant public IaaS platform.

2024 Examination (CSE-813, 54 marks)

Q5(a) — Two benefits and two drawbacks of cloud for enterprises [2.5 marks]

State two benefits and two drawbacks of cloud computing for enterprises.

Benefits:

1. **Reduced capital expenditure** — pay-as-you-go converts server buying (CAPEX) into a monthly utility bill (OPEX); small and medium enterprises access the infrastructure of corporate giants with no upfront investment.
2. **Elastic scalability on demand** — instantly provision extra compute for demand spikes (product launches, end-of-quarter runs) without buying hardware that sits idle the rest of the year.

Drawbacks:

1. **Security and privacy risk** — sensitive corporate data (IP, financials, customer records) resides on third-party shared infrastructure; data breaches or regulatory non-compliance (GDPR) expose the enterprise to liability.
2. **Vendor lock-in** — heavy use of provider-specific services (AWS Lambda, Azure Cosmos DB, Google BigQuery) makes migration to a competitor slow and expensive; the enterprise loses negotiating leverage over pricing and SLAs.

Q5(b) — Virtualization in cloud; two benefits to providers [2.5 marks]

Explain the fundamental concept of virtualization in cloud computing. State two benefits of virtualization to cloud providers for resource management.

Virtualization in cloud:

- A **hypervisor** (Virtual Machine Monitor) runs directly on physical hardware and **abstracts hardware resources** into multiple independent **Virtual Machines (VMs)**.
- Each VM has its own OS, CPU allocation, memory, and disk — fully **isolated** from other VMs on the same physical server.
- The stack: **Physical hardware** → **Hypervisor** → **VMs (Guest OS + Applications)**.
- Types: *Full virtualisation* (Type-1 bare-metal: VMware ESXi, Xen, KVM) and *para-virtualisation* (guest OS is aware of the hypervisor, lower overhead).

Two benefits to cloud providers:

1. **Dramatically higher server utilisation** — packing multiple VMs on one physical server raises utilisation from ~20% to >80%, reducing the number of physical machines (and their power/cooling cost) for a given workload.
2. **Rapid, flexible provisioning** — a new VM is created in seconds from a

template; the provider scales customer capacity instantly without shipping or installing new hardware, enabling true on-demand elasticity.

Q6(b) — IaaS vs. PaaS trade-offs for e-commerce migration [3 marks]

An e-commerce company is migrating to the cloud. Compare IaaS and PaaS on: development speed, operational overhead, vendor lock-in, and scalability.

Criterion	IaaS	PaaS
Dev speed	Slow — team must install and configure OS, web server, runtime, middleware	Fast — platform pre-configured; developer deploys code directly
Op. overhead	High — team manages OS patches, security hardening, scaling configuration	Low — provider manages OS, middleware, runtime updates
Vendor lock-in	Low — VMs are portable; standard OS (Linux/Windows) is the only dependency	High — proprietary runtimes, build APIs (e.g. AWS Elastic Beanstalk, Heroku)
Scalability	Manual — team configures auto-scaling rules and load balancers explicitly	Automatic — built-in horizontal scaling with no extra configuration
Best for	Database servers, legacy apps, custom network topologies	Stateless web/app tier, rapid prototyping, microservices

Recommendation for e-commerce: Use **PaaS for the application tier** (fast deployment, auto-scaling handles traffic spikes) and **IaaS for the database layer** (full control over DB configuration and storage volumes; lower lock-in risk for critical data).

Q6(c) — IaaS cost calculation: reserved vs. on-demand instances [3 marks]

Explain reserved instances vs. on-demand instances in IaaS pricing. Calculate the cost for a predictable base load and a variable peak workload.

Reserved instances: Pay an upfront commitment (1-year or 3-year term) in exchange for a **discounted hourly rate** (typically 30–70% cheaper than on-demand). Best for **predictable, always-on** workloads.

On-demand instances: Pay the full hourly rate with **no commitment**; terminate any time. Best for **variable, unpredictable** workloads, testing, or seasonal peaks.

Cost calculation example (monthly):

Workload	Detail	Monthly cost
Base load (reserved)	10 instances $\times$ 720 hr $\times$ \$0.05/hr	\$360
Peak load (on-demand)	5 extra instances $\times$ 100 hr $\times$ \$0.10/hr	\$50
Total		\$410

Decision rule:

- Instance runs $> 50\%$ of the time $\Rightarrow$ **reserved is cheaper**.
- Instance runs $< 50\%$ of the time $\Rightarrow$ **on-demand is cheaper**.
- Optimal strategy: **reserve the predictable base load**; use on-demand for traffic spikes.

Quick Summary — Exam Patterns & Must-Know

- **Two definitions to memorise:** NIST (“on-demand, shared pool, rapidly provisioned”) + Vaquero (“large pool of virtualised resources, pay-per-use, SLA”).
- **E-commerce deployment (2020 Q8b, 2021 Q3a, 2022 Q1b) — verbatim 3 years.** Lock in: IaaS (DB/storage) + PaaS (app tier) + SaaS (payment/email); Hybrid model (public storefront + private PII data). Prepare a 2-minute mental rehearsal.
- **Pros/cons (2020 Q8a, 2024 Q5a):** Pros = Flexibility, Scalability, Cost, Maintenance offloaded, Utilisation, Availability. Cons = Security, Privacy, Vendor lock-in, Network dependency, Migration.
- **Horizontal vs vertical scaling (2022 Q1a):** Vertical = one big machine; Horizontal = many small machines. Cloud favours horizontal. Know the 6-row comparison table.
- **Multi-tenancy (2022 Q1c):** Shared physical infra + logical isolation. Risks: data leakage, noisy neighbour.
- **Virtualization (2024 Q5b):** Physical $\rightarrow$ Hypervisor $\rightarrow$ VMs. Two provider benefits: utilisation + rapid provisioning.

- **IaaS vs PaaS (2024 Q6b):** Dev speed / overhead / lock-in / scalability — know the 4-row comparison table cold.
- **EUCALYPTUS (2021 Q5d):** Open-source private cloud, AWS-compatible API. Used when data cannot leave on-premises for security/compliance reasons.
- **Cloud vs Distributed (2021 Q5c):** Cloud IS a distributed system but adds pay-per-use, elasticity, multi-tenancy, provider-managed infrastructure.
- **EC2 motivation (2022 Q2a):** Amazon had excess capacity from holiday peaks — monetised it. Three VM benefits: utilisation + fine-grained billing + isolation.